Model of Big Bang Nucleosynthesis

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1 Introduction

The Big Bang created a large part of the lighter elements which still exist today in the Universe. In the first few milliseconds to seconds, it created a soup of extremely light particles, such as neutrinos, positrons or electrons. These light particles paired and unpaired to form neutrons and protons during a few milliseconds, while the temperature was rapidly dropping and the Universe expanding. In these first instant, protons and neutrons were rapidly exchanging particles, becoming alternatively one or the other. Once the temperature had fallen enough for them to stabilise, the neutrons and protons also started reacting together. These slightly heavier particles paired up into even larger particles, deuterium, tritium, helium, lithium and berillium. While most of the lithium and berillium decayed into helium, traces amount were still made in these times. This was the Big Bang nucleosynthesis. The heavier elements had to wait for stars to start their own nucleosynthesis to exist (Copi et al., 1995).

It is however not possible to observe this event directly. The only way we have, and might ever have, to understand the workings of this nucleosynthesis is by modeling it. Knowing the possible reactions between elements and their reaction rates today, especially at extremely high temperatures, we can make models of the way the Big Bang happened and what it created (Copi et al., 1995).

2 Methods

The reactions between elements which occured during the Big Bang can be modelled as a set of differential equations, in which positive terms are constructive, and negative ones are destructive. Each equation describes the evolution in time of one element, as a function of the rates of the different reactions, and the density of the elements involved in these reactions. This system can be solved using linear algebra. We will describe the algebraic process in part Derivation, and the reactions, reaction rates and the other variables of the system in Big Bang Nucleosynthesis.

2.1 Derivation

The equations in this set are

$$\frac{dn_j}{dt},$$

n_j being the number density of an element. We will call each of these terms $f_j(\vec{n})$, where j is the type of particle the equation concerns. Then,

$$f_j'(\vec{n_i})(\vec{n_{i+1}} - \vec{n_i}) = f_j(\vec{n_i}),$$

where i is a function of time, i+1 being the state of the equation after a step in time dt, from i. If we change this to concern a set of equations instead of a singular one, we obtain :

$$A'(\vec{n_i})(\vec{n_{i+1}} - \vec{n_i}) = A(\vec{n_i}).$$

A is a matrix, which contains the set of linear equations f_j at i, and $A'_j(\vec{n_i})$ depends on the Jacobian of the system, which is a matrix in which each term is

$$\frac{d}{dn_k}(\frac{\partial n_j}{\partial t}),$$

j and k can be equal. $A'_j(\vec{n_i})$ is such that :

$$A'(\vec{n_i}) = |I - dt \cdot J(n_i)|.$$

I is the identity matrix, of the same size as J, and dt is the time step taken between each points n_i. This is a system of linear equations, which can be solved after each times step dt, as:

$$A'(\vec{n_i})(\vec{n_{i+1}} - \vec{n_i}) = A(\vec{n_i}),$$

$$\Leftrightarrow \vec{n_{i+1}} = \vec{n_i} + \frac{A(\vec{n_i})}{A'(\vec{n_i})}.$$

For a better visualisation, A is the matrix :

$$A = \begin{bmatrix} \frac{dn_0}{dt} \\ . \\ . \\ . \\ \frac{dn_N}{dt} \end{bmatrix},$$

in which each function defines a different element in the system, and J, if we do not write the dt contribution, is the matrix:

$$\begin{bmatrix} \frac{dn_0}{dn_0} & \cdots & \frac{dn_0}{dn_N} \\ . & . & . \\ . & . & . \\ . & . & . \\ \frac{dn_N}{dn_0} & \cdots & \frac{dn_N}{dn_N} \end{bmatrix}$$

2.2 Big Bang Nucleosynthesis

Each equation in this system is of the shape :

$$\frac{dn_j}{dt} = \sum\sum n_l n_k \lambda_{lk} - n_j \sum n_k,$$

where positive terms are creative terms, which yield more of the element, and negative ones are destructive, changing the element into something else. The λs are the reaction rates, and indicate how much of each element is created/destroyed per time steps. We also transform these equations into mass fractions instead of densities, to avoid large numbers and make calculations easier. We now have :

$$\frac{dY_j}{dt} = \frac{1}{\rho N_A}(\sum\sum n_l n_k \lambda_{lk} - n_j \sum n_k),$$

where ρ is the total mass density in the system, and N_A is Avogadro's number $N_A = 6.02 \cdot 10^{23}\ mol^{-1}$. Because of this, the total density must be in mol/cm^{-3}, for the equations to yield the right units.

The reactions used in this model are these:

$$\begin{array}{ll}
(0) & p + n \to D + \gamma \\
(1) & D + D \to T + p \\
(2) & {}^3He + D \to {}^4He + p \\
(3) & T + D \to {}^4He + n \\
(4) & D + p \to {}^3He + \gamma \\
(5) & D + D \to {}^3He + n \\
(6) & {}^3He + n \to T + p
\end{array}$$

They were chosen as the most mentioned in literature referencing Big Bang nucleosynthesis, starting once the neutrons and protons had mostly stabilised, and not including elements heavier than helium (Baring, 2024; Navas et al., 2024; Ryan, 2008). We also include neutron decay, as :

$$(8) \qquad n \to p$$

Those reactions are sorted by most referenced, but the actual processus of creation and destruction would be better described in this way:

$$\begin{array}{llll}
n \to & p & & \\
p + n \to & D + \gamma & & \\
& D + D \to & T + p & \\
& D + p \to & {}^3He + \gamma & \\
& D + D \to & {}^3He + n & \\
& & {}^3He + n \to & T + p \\
& & {}^3He + D \to & {}^4He + p \\
& & T + D \to & {}^4He + n
\end{array}$$

To help with the computations, each particle was given an index (see table 1). Similarly, the reaction rates were given indexes, which are already indicated beside the reactions (eq. 2.2). Together, they formed a system of equations which describes this system. For example, the time derivative of the quantity of protons (particle 0) depends on five of the seven reactions, two destructive (0 and 4), and three constructive (1, 2 and 6) :

Particle	Index
p	0
n	1
D	2
γ	3
T	4
^3He	5
^4He	6

Table 1: Indexing of the elements represented in this system.

Reaction	Rate
(0)	$2.5 \cdot 10^4$
(1)	$3.9 \cdot 10^8 \cdot T_9^{-2/3} \cdot \exp(-4.26 \cdot T_9^{-1/3}) \cdot (1 + 0.0979 \cdot T_9^{1/3} + 0.642 \cdot T_9^{2/3} + 0.440 \cdot T_9)$
(2)	$2.6 \cdot 10^9 \cdot T_9^{-3/2} \cdot \exp(-2.99 \cdot T_9^{-1})$
(3)	$1.38 \cdot 10^9 \cdot T_9^{-3/2} \cdot \exp(-0.745 \cdot T_9^{-1})$
(4)	$2.23 \cdot 10^3 \cdot T_9^{-2/3} \cdot \exp(-3.72 \cdot T_9^{-1/3}) \cdot (1 + 0.112 \cdot T_9^{1/3} + 3.38 \cdot T_9^{2/3} + 2.65 \cdot T_9)$
(5)	$3.9 \cdot 10^8 \cdot T_9^{-2/3} \cdot \exp(-4.26 \cdot T_9^{-1/3}) \cdot (1 + 0.0979 \cdot T_9^{1/3} + 0.642 \cdot T_9^{2/3} + 0.440 \cdot T_9)$
(6)	$7.06 \cdot 10^8$
(7)	$1/800$

Table 2: Reaction rates in the system

$$
\begin{aligned}
\frac{dY_0}{dt} &= \frac{1}{\rho N_A}(-\lambda_0 n_0 n_1 + \lambda_1 n_2^2 + \lambda_2 n_5 n_2 + \lambda_6 n_5 n_1 - \lambda_4 n_2 n_0 + \lambda_9 n_2) \\
\frac{dY_1}{dt} &= \frac{1}{\rho N_A}(-\lambda_0 n_0 n_1 + \lambda_3 n_4 n_2 + \lambda_5 n_2^2 - \lambda_6 n_5 n_1 - \lambda_9 n_2) \\
\frac{dY_2}{dt} &= \frac{1}{\rho N_A}(\lambda_0 n_0 n_1 - \lambda_1 n_2^2 - \lambda_2 n_5 n_2 - \lambda_3 n_4 n_2 - \lambda_4 n_2 n_0 - \lambda_5 n_2^2) \\
\frac{dY_3}{dt} &= \frac{1}{\rho N_A}(\lambda_0 n_0 n_1 + \lambda_4 n_2 n_0) \\
\frac{dY_4}{dt} &= \frac{1}{\rho N_A}(\lambda_1 n_2^2 - \lambda_3 n_4 n_2 + \lambda_6 n_5 n_1) \\
\frac{dY_5}{dt} &= \frac{1}{\rho N_A}(-\lambda_2 n_5 n_2 + \lambda_5 n_2^2 + \lambda_4 n_2 n_0 - \lambda_6 n_5 n_1) \\
\frac{dY_6}{dt} &= \frac{1}{\rho N_A}(\lambda_2 n_5 n_2 + \lambda_3 n_4 n_2)
\end{aligned}
\tag{1}
$$

As the Jacobian of the system is derived from this, and it is quite a long derivation, it can be found in Appendix A. Finally, the reaction rates and initial densities are the only elements missing from this system. The reaction rates are taken from Wagoner et al., 1967, except for the neutron decay into protons, which is taken as the inverse of its half-life, about 888 seconds (Navas et al., 2024).

Those rates depend on the temperature of the system and on the total density. At this time, temperature decreased with time, as shown in eq. 2. It also varied with the inverse of the expansion rate of the Universe $T \propto R(t)^-1$ (Kolb, 1990).

$$T(t) = \frac{1.5 \cdot 10^{10}}{t^{1/2}} \quad (2)$$

The time range was from t = 1s to $t = 6 \cdot 10^5 s$. This is more than the time it took to form helium according to current models, but this is about the time it took for our model to stabilise. An end closer to a 100s would be better, as that corresponds to the time when the amount of helium mostly stabilised (Kolb, 1990). The reasons why this might not have worked in this model are discussed in Discussion and Conclusion.

The last element missing to run this model is the initial densities. As they are not yet formed at t=0.1, we can already know that the number densities of anything larger than the neutrons and protons will be 0. The only exception could be the γ particles, but as they are not destroyed, only created here, we can still set it to 0 without affecting our system. Remains the proton and neutron density. We set them to be $n_0 = 0.85$ and $n_1 = 0.15$ of the total density, as the ratio of proton to neutron was about seven to one (Schramm & Turner, 1998). The total density was about $2 \cdot 10^{-31} g/cm^3$ (Copi et al., 1995), which, as the molar mass of neutron and proton is about one, gives us $\rho = 2 \cdot 10^{-31}$.

3 Results

3.1 Simulations

We can see the results of the simulation on figure 1. On it, we can observe that the temperature drops low enough for the nucleosynthesis to start around 5s after the Big Bang. We can also observe when different reactions started happening or became more effective than others. The dip in T is probably due to its reaction with D to form 4-helium. The one in D is both form this reaction and the ones it has with itself to form T and 3-helium. The amount of γ grows very fast and never drops because it is only created, not destroyed. This is also the case for 4-helium.

Comparing our results to those

3.2 Solving the system differently

In this model, we solved the equation system by deriving the Jacobian by hand, analytically, then solving the system using matrix inversion. It is also possible to find the Jacobian at each time step numerically. Both methods have their uses in different situations.

The analytical method is exact, there is no errors in the values calculated at any points, but it can not be used for every system of equations. Sometimes the functions are not derivable, and it is then not possible to solve them analytically. It is also very demanding ahead of the computation, and the errors can slip in not as an uncertainty in the values but as missing factors, wrong numbers and such. This system was already composed of 7×7 derivatives, and that number scales with the number of elements involved, so the derivatives themselves get longer, and thus the risk of an analytical error increases with each new reaction. A system such as the one used in Wagoner et al., 1967, which has more than eighty reactions and their reverse, is so large that solving it analytically would require a lot of work.

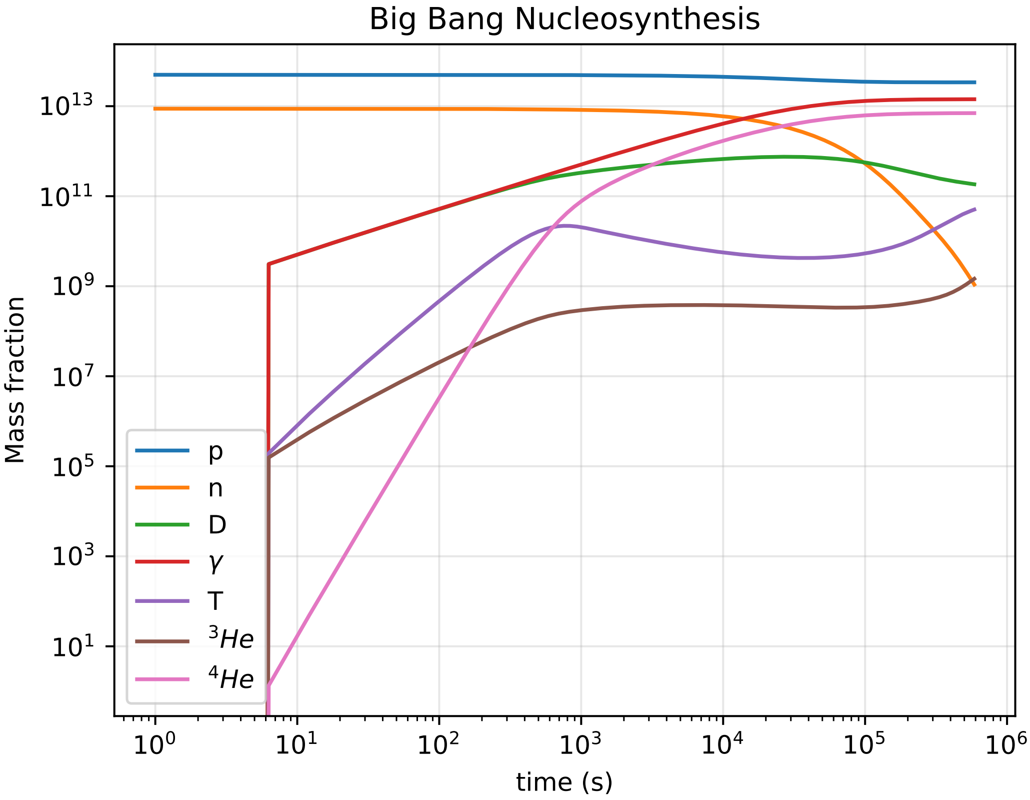

Figure 1: Figure showing the evolution of the abundance of different elements in the model across time. The abundances are shown as a number density, and not a mass fraction as the label states.

On the other hands, numerical solutions are not exact. They take a small step Δx in a direction and find an approximation of the analytical function for the value after that step. If it is too much of an approximation, or if the step is too large, the uncertainty also grows. This can lead to extremely large errors in the final results. However, as the entire set of functions only needs to be inputed in a program which does the calculations automatically, if the program is well made and controls the step and approximation to reduce the error, it could theoretically compute an infinite amount of reactions. In the case of the modelisation of astrophysical phenomena, numerical solution also allow for more realistic conditions. Idealised conditions, such as thermodynamic equilibrium, can be solved analytically, but as soon as the system is no longer ideal, numerical solutions might become necessary. Another problem with numerical solution is that they are much more demanding in terms of computation time and power. As it requires an iteration to find a solution at each step, a large set of equations and large amount of steps can take very long. In some cases, long enough that deriving the analytical solution and plugging this into the machine is a better solution.

In the case of this system, with a relatively small amount of equations, all of them derivable, but also quite simple to solve numerically, both solutions work well. The analytical one requires slightly more work ahead of the computation to find the

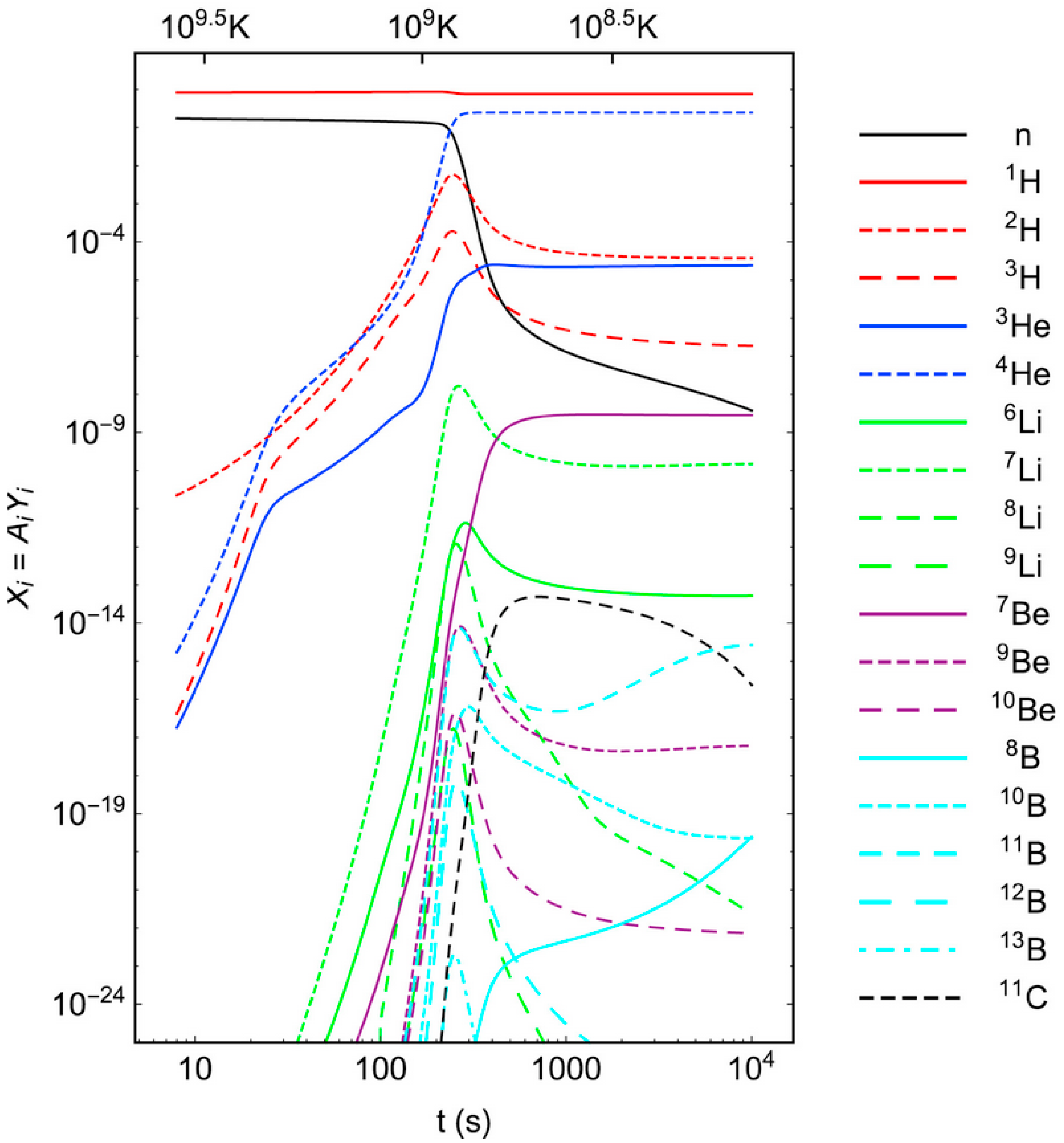

Figure 2: Figure showing the evolution of the abundance of the elements included in the high precision model of Big Bang Nucleosynthesis from Pitrou et al., 2018.

derivatives, but the computation itself is easier and not very demanding. The numerical requires less work ahead, but for the amount of steps we took (sometimes more than a million), it might take much longer than the analytical to compute.

Here, we are mixing both, by solving the Jacobian analytically, and then solving the system of differential equations numerically at each time steps, using the implicit Euler method and solving this new system using matrix inversion.

3.3 Adaptative step size

When a numerical solution is used, one must choose a step size after which the function is re-evaluated. To resolve the system to the best resolution, a constant step size could be a problem. If it is too large, it risks missing important features or creating large errors. On the contrary, if it is too small, it risks looping eternally on slowly evolving features. An adaptative step size which depends on the rate of change of the system seems to be the best option. Instead of directly evaluating the function at a new point, the machine could first measure how "far away" it is from the previous step, and reduce the step size before doing its evaluation. This step size can also be linked to the error, by setting it to a large number, and then iterating until the error is smaller than some set value. An adaptive step size requires more computer power, as it adds to the iterations, but using a language such as fortran, it remains quite reasonable.

4 Discussion and Conclusion

While this model represents the bare bones of the first part of the Big Bang nucleosynthesis, the difference shown between other models, this model, and observed densities can be explained by the simplifications made here. In particular, a number of reactions were not included. Wagoner et al., 1967 uses more than eighty reactions and their reverse, for example. We also do not model any reverse reaction, which influences the densities we find.

One phenomenum which is important in Big Bang nucleosynthesis but was not included here is the expansion, which reduces density. We know the dependence of the expansion rate of the Universe on time, the relation between temperature and time, and the relation between temperature and the expansion rate. We also know that $\rho \propto 1/V$ and $V \propto R^3$. Using the equation we have for the temperature, we can set :

$$R(t) = \frac{\sqrt{t}}{4 \cdot 10^{10}}$$

$$\rho(t) = \frac{3\rho_0}{4\pi R(t)}$$

The code used and a LAtex version of this report can be found at , in "Project"

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5 Appendices

5.1 Appendix A

The Jacobian is composed of 7×7 functions, each term being the derivative of $\frac{dn_i}{dt}$ as a function of n_j, where i can be equal to j.

The terms in the first row are :

$$\begin{aligned}
J_{00} &= -\lambda_0 n_1 - \lambda_4 n_2 \\
J_{01} &= -\lambda_0 n_0 + \lambda_6 n_5 \\
J_{02} &= 2\lambda_1 n_2 + \lambda_2 n_5 - \lambda_4 n_0 \\
J_{03} &= 0 \\
J_{04} &= 0 \\
J_{05} &= \lambda_2 n_2 + \lambda_6 n_1 \\
J_{06} &= 0
\end{aligned}$$

Second row:

$$\begin{aligned}
J_{10} &= -\lambda_0 n_1 \\
J_{11} &= -\lambda_0 n_0 - \lambda_6 n_5 \\
J_{12} &= \lambda_3 n_4 + 2\lambda_5 n_2 \\
J_{13} &= 0 \\
J_{14} &= \lambda_3 n_2 \\
J_{15} &= -\lambda_6 n_1 \\
J_{16} &= 0
\end{aligned}$$

Third row:

$$J_{20} = \lambda_0 n_1 - \lambda_4 n_2$$
$$J_{21} = \lambda_0 n_0$$
$$J_{22} = -2\lambda_1 n_2 - \lambda_2 n_5 - \lambda_3 n_4 - \lambda_4 n_0 - 2\lambda_5 n_2$$
$$J_{23} = 0$$
$$J_{24} = -\lambda_3 n_2$$
$$J_{25} = -\lambda_2 n_2$$
$$J_{26} = 0$$

Fourth row:

$$J_{30} = \lambda_0 n_1 + \lambda_4 n_2$$
$$J_{31} = \lambda_0 n_0$$
$$J_{32} = \lambda_4 n_0$$
$$J_{33} = 0$$
$$J_{34} = 0$$
$$J_{35} = 0$$
$$J_{36} = 0$$

Fifth row:

$$J_{40} = 0$$
$$J_{41} = \lambda_6 n_5$$
$$J_{42} = 2\lambda_1 n_2 - \lambda_3 n_4$$
$$J_{43} = 0$$
$$J_{44} = -\lambda_3 n_2$$
$$J_{45} = \lambda_6 n_1$$
$$J_{46} = 0$$

Sixth row:

$$J_{50} = \lambda_4 n_2$$
$$J_{51} = -\lambda_6 n_5$$
$$J_{52} = -\lambda_2 n_5 + 2\lambda_5 n_2 + \lambda_4 n_0$$
$$J_{53} = 0$$
$$J_{54} = 0$$
$$J_{55} = -\lambda_2 n_2 - \lambda_6 n_1$$
$$J_{56} = 0$$

Seventh row:

$$
\begin{aligned}
J_{60} &= 0 \\
J_{61} &= 0 \\
J_{62} &= \lambda_2 n_5 + \lambda_3 n_4 \\
J_{63} &= 0 \\
J_{64} &= \lambda_3 n_2 \\
J_{65} &= \lambda_2 n_2 \\
J_{66} &= 0
\end{aligned}
$$